UNIVERSITY OF NIAGARA FALLS CANADA

MASTER OF DATA ANALYTICS

CREDIT RISK ANALYSIS AND MODELING USING THE UCI CREDIT CARD DEFAULT DATASET

PREPARED BY

SANDESH DHAKAL

SUBMITTED TO

PROF. TOURAJ BANIROSTAM

June 13, 2025

Table of Contents

Table of Figures

Special instructions for running this Power BI file on other computers:

- Adjust the screen size to 70% for better visualization and viewing.
- Advanced visualization and regression modeling were done using Python. So, please download the Python environment and install the following libraries: Matplotlib and scikit-learn (sklearn).

1. Introduction

For our group project, we worked with the UCI Credit Card Default Dataset. This is a big set of information about more than 30,000 credit card customers. Our main goal was to understand which types of customers are most likely to miss their payments (which is called "defaulting") and how banks can use these patterns to make better decisions. We used Power BI to make an interactive dashboard, and we also used some simple AI methods to help predict defaults.

2. Data Exploration and Preparation

The provided dataset includes the following key attributes:

- **Demographics**: AGE, SEX, EDUCATION, MARRIAGE
- **Credit Behavior**: LIMIT_BAL (credit limit), BILL_AMTn (monthly bill amounts), PAY_AMTn (monthly payments), PAY_n (past due payment status)
- **Target Variable**: DefaultNextMonth (1 = Default, 0 = No Default)

We started by loading the dataset into Power BI. We took a good look at all the columns, like age, gender, education level, marriage status, credit limit, monthly bills, and payment history. Some of the column names were confusing, so we changed them to make everything clearer. For example, we changed "SEX" to "Gender" and "EDUCATION" to "Education Level." If any data was missing or didn't look right, we removed those rows so our results would be accurate. We also looked for numbers that were very different from the rest (called outliers) and fixed those. To get even more useful information, we created new columns, like the total bill amount and total payment for each person, and we put customers into age groups.

3. Exploratory Data Analysis (EDA)

After cleaning the data, we explored the basic patterns. We checked if some groups of people were more likely to default than others. For example, we looked at gender, education, marriage status, and age. We found that men, single people, and those with only a high school education are more likely to miss payments. We also saw that younger customers (in their 20s) and older customers (over 50) had higher default rates than people in the middle age ranges. This early analysis helped us focus on which groups might need more attention.

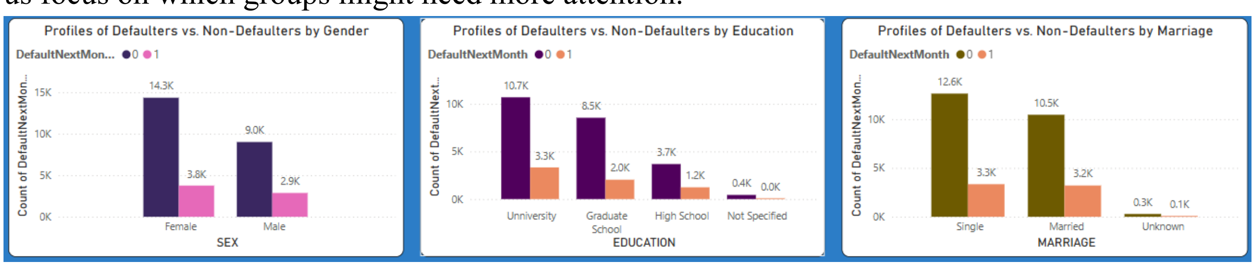

Figure 1: summary of customer demographics, payment behaviors, and credit usage

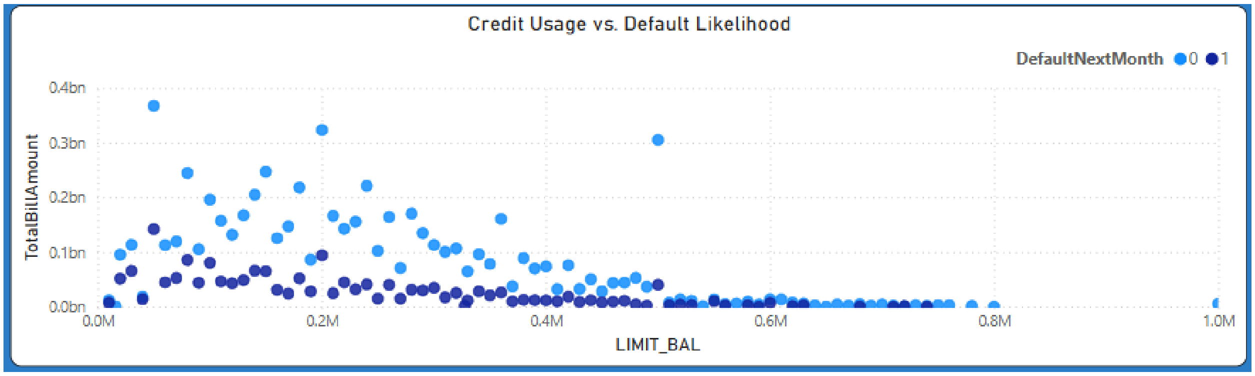

Figure 2: Credit Utilization Vs Default Likelihood

4. Interactive Reports and Dashboards

Next, we created our interactive Power BI dashboard. At the top of the dashboard, we added filters (called "slicers") so anyone can focus on specific groups, like only women, only married people, or certain age ranges. There are bar charts showing defaulters by gender, education level, and marriage status, making it easy to compare groups. The big blue histogram in the middle shows how defaults are spread across different ages. We also included a scatter plot ("Credit Usage vs. Default Likelihood") to show that people who use a lot of their credit are more likely to default. On the left, the "Key Influencers" chart clearly shows that having a high bill amount is a strong sign that someone might default. These visuals make it easy to spot patterns, even for someone new to data analysis.

Figure 3: Credit Risk Visualization and Advanced Analysis Dashboard

5. Key Insights and Model Outcomes

We wanted to take things a step further, so we used some simple AI tools. We did Principal Component Analysis (PCA), which helped us see which features (like bill amount, payments, and credit limit) mattered most for grouping customers by risk. We noticed two main groups: a high-risk group and a low-risk group. Then, we built a logistic regression model in Python. This model was able to predict if someone would default about 77% of the time. It worked best at showing who was not likely to default, but it also gave us useful information about people who are at risk.

Python Code for Logistic Regression:

```
# Prepare the dataset
X = dataset[['LIMIT_BAL', 'AGE', 'BILL_AMT1', 'PAY_AMT1']]  # Features
y = dataset['DefaultNextMonth']  # Target variable

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train the model
model = LogisticRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred)

# Output the accuracy and classification report to Power BI
output = pd.DataFrame({
    "Accuracy": [accuracy],
    "Classification Report": [report]
})

# Plotting accuracy as a simple text
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(6, 3))
ax.text(0.5, 0.5, f"Accuracy: {accuracy:.2f}", ha='center', va='center', fontsize=14)
ax.axis('off')
plt.show()
```

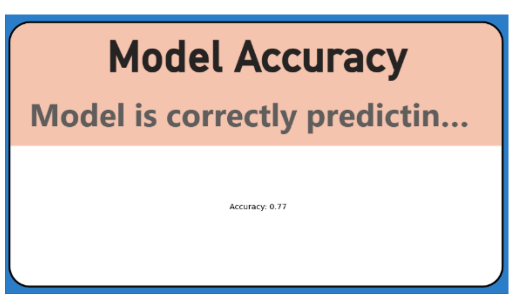

Figure 4: Model Accuracy

6. Interpretation and Business Implications

Our main takeaway is that customers with high credit limits but who make low payments are more likely to default, so banks should watch these customers closely. Younger and single customers also default more often, so it might help banks to offer them financial advice or support. Our analysis shows that things like bill amount, payment history, and credit limit are very important for predicting risk. By using these insights, banks can offer better, more personalized services and reduce their own risk.

7. Conclusion

Working as a group, we explored, cleaned, and analyzed the data, built an AI model, and put everything together in a user-friendly Power BI dashboard. Our project shows that with a mix of data analysis, simple machine learning, and clear visuals, even a big problem like credit risk can be understood and explained. This approach can really help banks and other financial companies make smarter choices and better support their customers.